Vehicular bottom-up emissions inventory and atmospheric simulation for 58 urban centers of South America

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Introduction

An emission inventory is an important tool for environmental policy. Emissions from vehicles are difficult to estimate due to complexity in identify and quantify sources in movement. For Southamerica, only few cities have reported emissions inventory at street level (Bottom-up), like Santiago[1], Bogotá[2], Buenos Aires[3] and Brasilia[4] and a standardized procedure is needed.

Objective

Develop a standardized method for estimate vehicle emissions in Southamerican Cities at street level

Method

Regressions of traffic counts based on type of street and distance from down-town using the R package remIAG0.3[5]. Traffic counts comes from São Paulo. Emission factors from FTP75 tests divided for Copert[6] emission factors at 34.12 km/h. It was considered the effect bio fuels on Brazil, and without on other countries. It was associated the average age of passenger cars with GDPpp of year City.

$$E_{ijklm} = F_i * AF_n * TV_k * L_i * EF(V_o)_{jk} * DP_{lm} \quad [5]$$

$$EF(V)_{i,j,k,l} = EF(V)^{Copert}_{i,j,k} * REF_{i,j,k} * DF_{i,j,k,l} \quad [5]$$

$$S_A = 1 - \exp(-\exp(1.798 - 0.137t)) \quad [6]$$

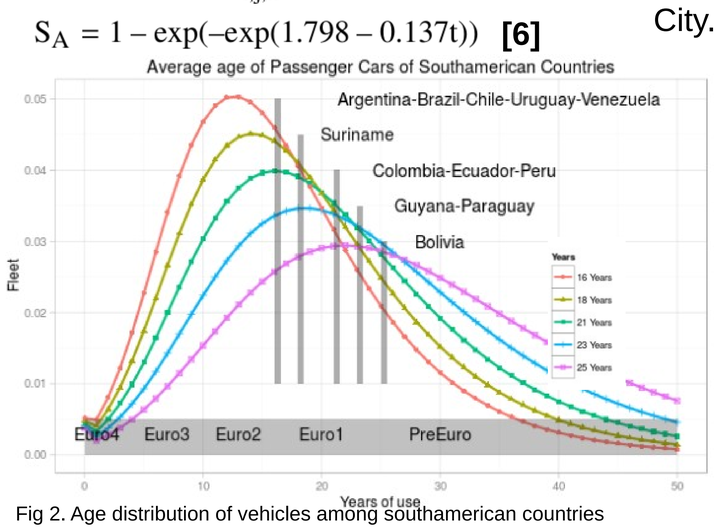


Fig 2. Age distribution of vehicles among southamerican countries

Results

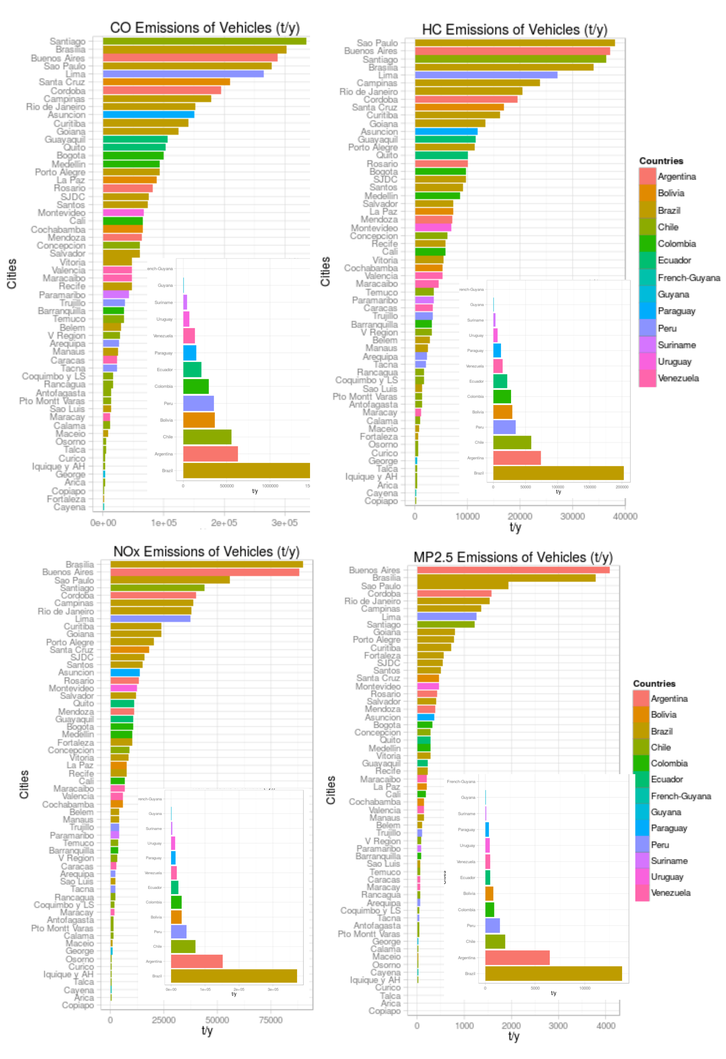
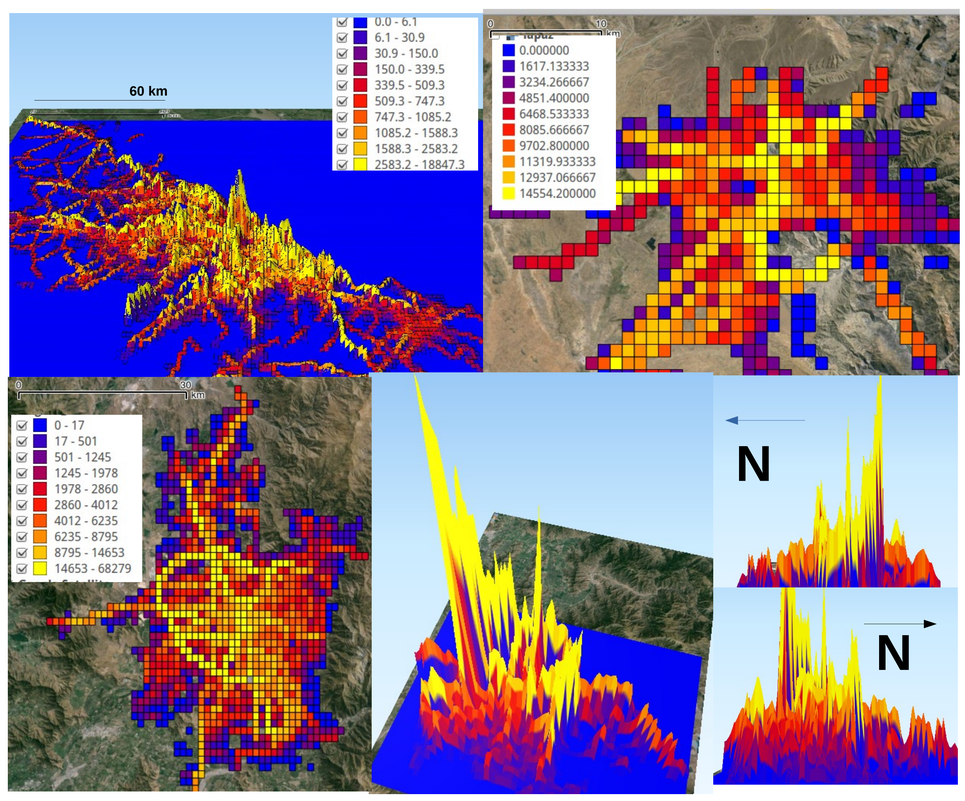


Fig 3.Vehicular emissions CO, HC, NOx and MP2.5 for 2012 (t/y)



Fig; 1. CO emissions from Buenos Aires, Argentina, La Paz, Bolivia and Santiago, Chile (mol CO/km/h)

Numerical Experiment

It was performed numerical modelling of CO for the city of Santiago for the periods from 6 June of 2012 to 8 June 2012 for test. In that period it the city was declared environmental emergency due high levels of air pollutant. The model was WRF Chem 3.6 [7] with the following configuration. Despite that it was used GFS 1 degree as initial conditions, The results shows good agreement with stations located in south of the City.

Atmospheric Process	Wrf-Chem option
Longwave radiation	RRTM
Shortwave radiation	Goddard
Surface layer	Monin-Obukhov
Land surface	Noah
Boundary layer	YSU
Cumulus clouds	Grell 3D
Cloud microphysics	Lin
Gas-phase chemistry	RADM2
Photolysis	Fast-J

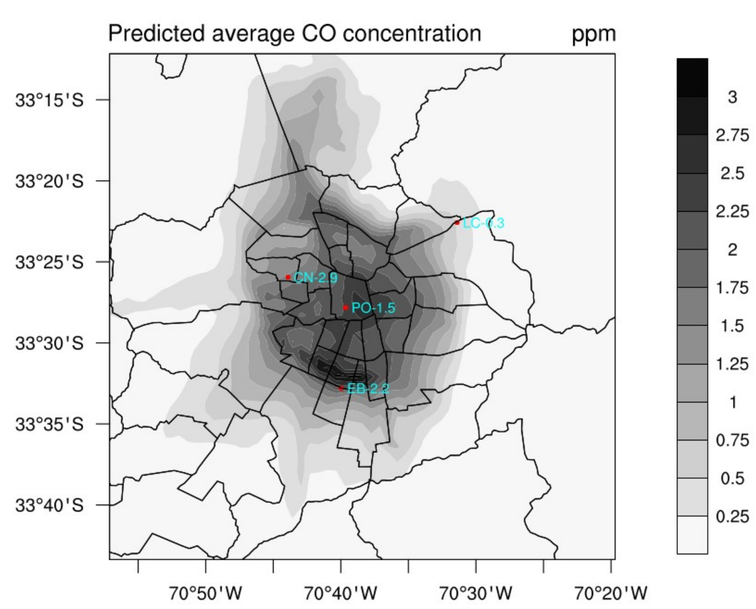


Fig 4.Simulated CO averaged for 6,7 and 8 of June, 2012

Discussion/Conclusions

Comparing Buenos Aires 2012 and 2006[3], CO -49%, NOx +8%, HC -46%, MP -35%. Comparing Santiago 2012 and 2010 projected[8], CO +125%, NOx +15%, HC +194%, MP +3%. Comparing Brasilia 2012 and 2010[4], CO +154%, NOx -41%, HC -80%, MP -25%.

NOx and MP2.5 shows the best agreements. Nevertheless, differences are due to roadlength, emission factors and approach used, average age of vehicles, as example.

Table 1. Vehicular emissions of MP2.5, CO, Nox, HC (t/y) and road-length (km) for year 2012.

ID	Countries	Cities	MP2.5	CO	NOx	HC	longkm
1	Argentina	Buenos Aires	4093	288577	88283	37167	40359
2	Argentina	Cordoba	1577	195149	40099	19485	8802
3	Argentina	Mendoza	388	63756	10911	7060	3442
4	Argentina	Rosario	431	82657	13125	10014	4441
5	Bolivia	Cochabamba	153	65944	5728	5277	2484
6	Bolivia	La Paz	200	88960	7760	7209	3955
7	Bolivia	Santa Cruz	457	208934	18039	16906	5304
8	Brazil	Belem	112	30381	4237	2872	2359
9	Brazil	Brasilia	3797	303119	90092	34029	11696
10	Brazil	Campinas	1354	179127	38830	23850	10676
11	Brazil	Curitiba	723	141631	23987	16233	6833
12	Brazil	Fortaleza	559	1802	10009	622	5275
13	Brazil	Goiana	805	124970	23952	13487	9366
14	Brazil	Maceio	25	7801	1008	780	1213
15	Brazil	Manaus	139	25404	4201	2488	3013
16	Brazil	Porto Alegre	778	93098	20297	11320	7972
17	Brazil	Recife	224	47599	7624	5883	3772
18	Brazil	Rio de Janeiro	1529	151983	38014	20436	18260
19	Brazil	Salvador	397	60233	11927	7251	3438
20	Brazil	Santos	495	74561	15094	9044	3165
21	Brazil	Sao Luis	71	12671	2190	1330	1603
22	Brazil	SJDC	534	75876	15691	9682	3735
23	Brazil	Sao Paulo	1931	278446	55644	38142	26440
24	Brazil	Vitoria	276	48413	8327	5473	3658
25	Chile	Antofagasta	47	13200	1622	1302	461
26	Chile	Arica	11	3479	411	331	186
27	Chile	Calama	30	11438	1229	1019	323
28	Chile	Concepcion	283	61404	9053	6194	2501
29	Chile	Copiapo	9	2126	272	196	127
30	Chile	Coquimbo y LS	54	17210	2016	1667	875
31	Chile	Curico	15	4371	531	527	230
32	Chile	Iquique y AH	16	4033	517	354	283
33	Chile	Osorno	21	5877	720	614	367
34	Chile	Pto Montt Varas	39	12677	1507	1328	769
35	Chile	Rancagua	61	17084	2132	1683	598
36	Chile	Santiago	1221	335777	43979	36393	11191
37	Chile	Talca	14	4478	487	427	533
38	Chile	Temuco	69	33923	3829	3596	733
39	Chile	V Region	92	27818	3384	3132	1790
40	Colombia	Barranquilla	87	34987	3418	3261	1536
41	Colombia	Bogota	328	99742	10668	9694	6291
42	Colombia	Cali	188	66275	6763	5772	1898
43	Colombia	Medellin	279	93916	10107	8628	1977
44	Ecuador	Guayaquil	224	107106	10671	11475	3867
45	Ecuador	Quito	280	103011	11008	10149	3809
46	Guyana	George	33	3646	909	433	535
47	F-Guyana	Cayena	16	1633	425	204	268
48	Paraguay	Asuncion	359	150260	13542	11948	4967
49	Peru	Arequipa	58	26544	2394	2286	2334
50	Peru	Lima	1255	264943	37166	27089	11808
51	Peru	Tacna	50	22743	2165	2005	724
52	Peru	Trujillo	110	36857	4058	3311	1464
53	Suriname	Paramaribo	90	42952	4014	3496	931
54	Uruguay	Montevideo	456	67572	12300	6831	5845
55	Venezuela	Caracas	68	23873	2578	3366	1170
56	Venezuela	Maracaibo	206	47797	6672	4576	2030
57	Venezuela	Maracay	62	12285	1883	1158	1122
58	Venezuela	Valencia	149	47831	5935	5261	1734
	Southamerica	**Total**	**27329**	**4459961**	**773432**	**485745**	**266537**

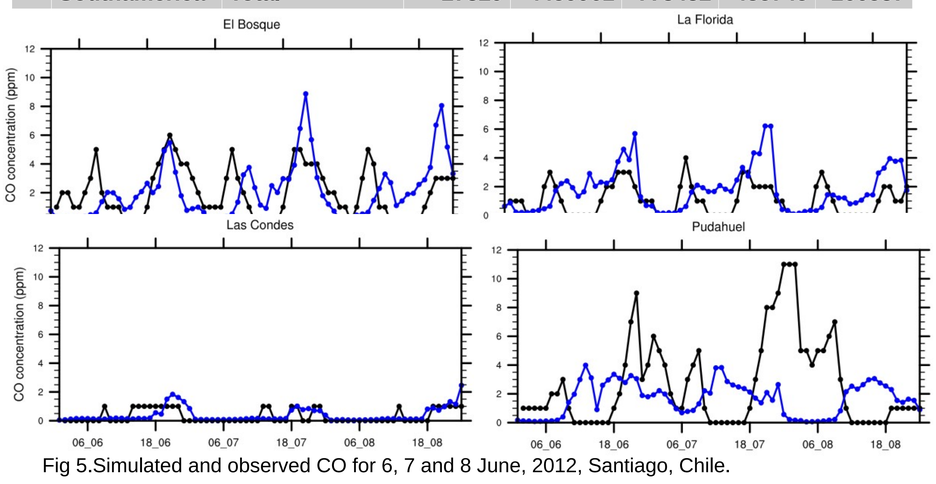


Fig 5.Simulated and observed CO for 6, 7 and 8 June, 2012, Santiago, Chile.

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